

SAKSHAM GUPTA

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EDUCATION

Manipal Institute of Technology, Manipal

Manipal, Karnataka • May 2027

B.Tech in Information Technology

CGPA: 9.16/10.0

Delhi Public School R.K. Puram

New Delhi • 2023

Class 12 – CBSE (90.2%)

Class 10 – CBSE (93.5%)

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, C++, C

Computer Science Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems

Web Technologies: Flask, Node.js, React, REST APIs, HTML/CSS

Software Engineering: SDLC, Agile Methodology, Requirements Engineering, Software Testing, UML Diagrams, Risk Management

Databases & Tools: SQL (MySQL, PostgreSQL), MongoDB, Git, GitHub Actions (CI/CD), Docker

Artificial Intelligence & Machine Learning: Pandas, NumPy, Scikit-learn, Visualization (Matplotlib, Seaborn), Data Analysis

Systems, Security & Tools: Linux, Bash/Shell Scripting, Multithreading & Concurrency Control, AES-256 Encryption, Digital Signatures, Secure Key Management

EXPERIENCE

Software Development Intern

June – July, 2025

Indian Railway Catering and Tourism Corporation (IRCTC)

- Analyzed enterprise booking system architecture handling 2M+ daily transactions, identifying data flow and performance bottlenecks
- Optimized SQL query execution plans and indexing strategies, reducing response latency by 25% in high-volume transaction environment
- Evaluated scalable system design patterns including microservices architecture for high-traffic applications
- Designed Python-based load testing scripts to simulate concurrent transactions, analyzing latency and system throughput under high traffic

PROJECTS

SafePay | High-Throughput Transaction Fraud Detection System | [GitHub](#) • [View Live](#)

Python, Flask, MongoDB, Scikit-learn, Cryptography

- Developed a secure Flask-based payment system integrating ML-driven fraud detection (85%+ accuracy) with real-time transaction processing
- Implemented AES-256 encryption and digital signatures to ensure confidentiality and integrity of transaction data
- Secured authentication using password hashing and enforced Role-Based Access Control (RBAC) with concurrency-safe balance updates to prevent race conditions

Fileshare | Real-Time Anonymous File & Chat Sharing | [GitHub](#) • [View Live](#)

Node.js, Express, Socket.io, React, MongoDB, HTML, CSS, JavaScript

- Built a real-time anonymous room-based file and chat sharing web application using Node.js, Express, and Socket.io
- Developed ephemeral rooms with automatic expiry to ensure privacy and minimal data retention
- Enabled real-time file uploads (≤10MB) using Multer with cloud storage integration and deployed the full-stack application to cloud

Starfinder | Multi-Class Stellar Classification System | [GitHub](#) • [View Live](#)

Python, Flask, Scikit-learn, Pandas, NumPy, JavaScript

- Developed multi-class stellar classification using Random Forest, achieving 93% accuracy across 6 star types
- Built input validation to handle out-of-distribution data and maintain prediction reliability
- Deployed a production ML system with Flask REST API, automated model persistence, and data analysis pipeline

Code2Flow | AI Code-to-Flowchart Generator | [GitHub](#) • [View Live](#)

Python, Flask, JavaScript, HTML, CSS

- Built an AI-powered tool that converts source code into deterministic ASCII-based flowcharts using Flask and Gemini API
- Designed a pipeline separating control-flow extraction from deterministic rendering for consistent output
- Established normalization and validation to handle LLM formatting variations and minimize API calls

LEADERSHIP & ACTIVITIES

Organizing Committee Member — Sports (Chess)

March 2025

Revels'25, MIT Manipal

- Led logistics coordination for chess tournaments with 200+ participants, managing registration and bracket systems
- Collaborated with a 12-member cross-functional team ensuring 100% on-time event execution during cultural fest

ACHIEVEMENTS

LeetCode: Solved 200+ problems across data structures and algorithms

Achievers Scholarship: Ranked in Top 5% of Information Technology cohort

Academic Excellence: Maintained 9.0+ GPA across all semesters at MIT Manipal

ADDITIONAL

Languages: Hindi (Native), English (Proficient)

Interests: Competitive Programming, Chess, Badminton, Reading